

机器学习度量

Machine Learning Measures

明玉瑞 Yurui Ming
yrming@gmail.com

声明

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例子

Example

- ▶ 假设我们要学习某个类别（概念） C ，以跑车为例：
 - ▶ 给定一组汽车样本，由人工将其标注为跑车（正样本）或非跑车（负样本）；
 - ▶ 任务：找到一个描述规则，使其适用于所有正样本，且不适用于任何负样本。
- ▶ 当得到了类别 C 的定义后，可以实现以下两点：
 - ▶ 预测——给定一辆从未见过的新车，判断它是否属于跑车
 - ▶ 描述/信息压缩——理解人们对于跑车这类车型所怀有的预期
- ▶ Suppose we want to learn a class (concept) C , exemplified by sports car:
 - ▶ given a collection of cars, have people label them as sports car (positive example) or non-sports car (negative example)
 - ▶ task: find a description that is shared by all of the positive examples and none of the negative examples
- ▶ Once we have this definition for C , we can
 - ▶ predict – given a new unseen car, predict whether or not it is a sports car
 - ▶ describe/compress – understand what people expect in a car

建模

Modeling

- 为了简化问题处理流程，排除无关属性的干扰，假设在描述汽车的所有特征中，我们选取价格与发动机功率这两项，现定义：

- x_1 : 代表汽车价格

- x_2 : 代表发动机排量

- 在此前提下，每辆车可以表示为 $\mathbf{x} = [x_1, x_2]$ ，我们用标签 y 表示车辆的类型，即：

$$y = \begin{cases} 1 & \text{车为赛车} \\ 0 & \text{车非赛车} \end{cases}$$

- To make things simpler and allow us to ignore irrelevant attributes, we choose price and engine power among all the features describing cars. Define:

- x_1 : represent the price

- x_2 : represent the engine volume

- Then each car is represented as $\mathbf{x} = [x_1, x_2]$. And the label y denotes its type:

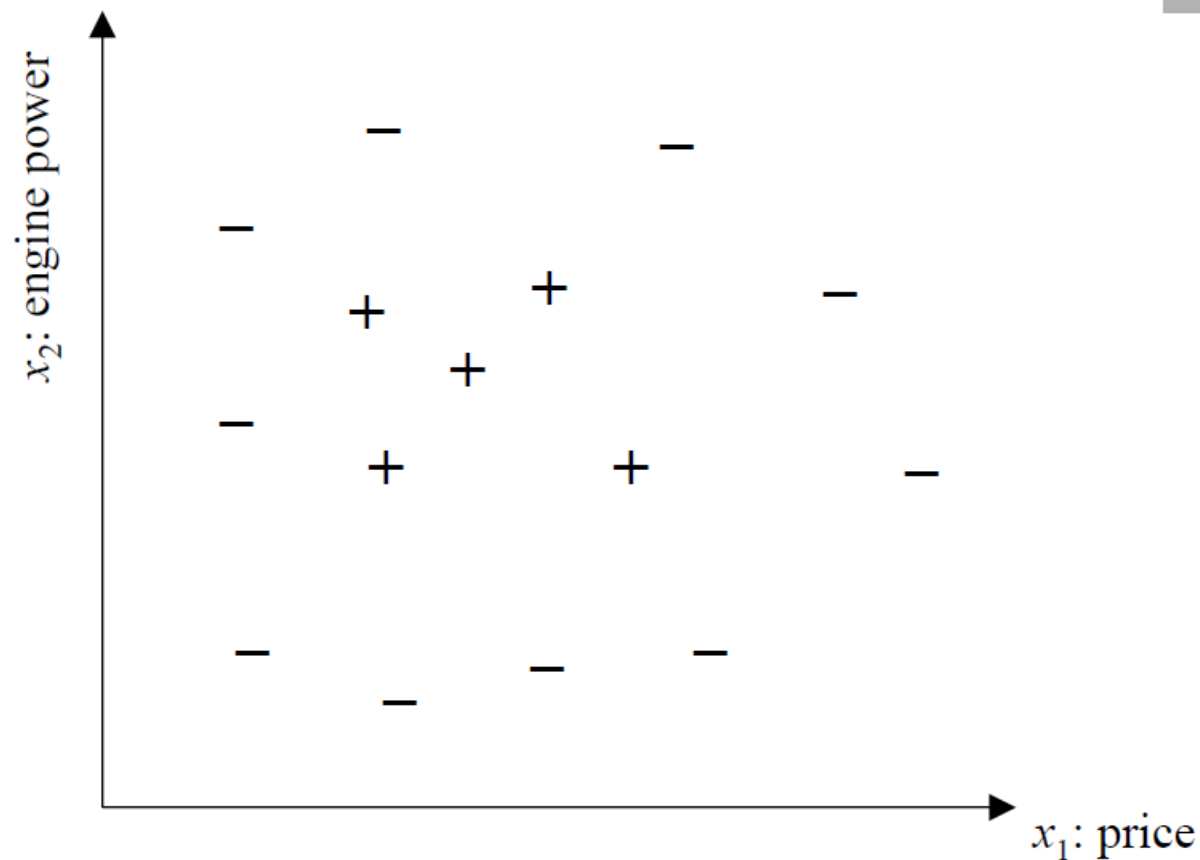
$$y = \begin{cases} 1 & \text{sport car} \\ 0 & \text{nonsport car} \end{cases}$$

建模

Modeling

- 根据上面定义，每个样本可以由元组 $(\mathbf{x}, y)$ 表示，所有训练样本的集合，其数量记为 N ，可以表示为 $\mathcal{X} = \{(\mathbf{x}^t, y^t)\}_{t=1}^N$ 。

Based on the above assumption, each example is represented by the pair $(\mathbf{x}, y)$, and a training set containing N examples is represented by $\mathcal{X} = \{(\mathbf{x}^t, y^t)\}_{t=1}^N$.



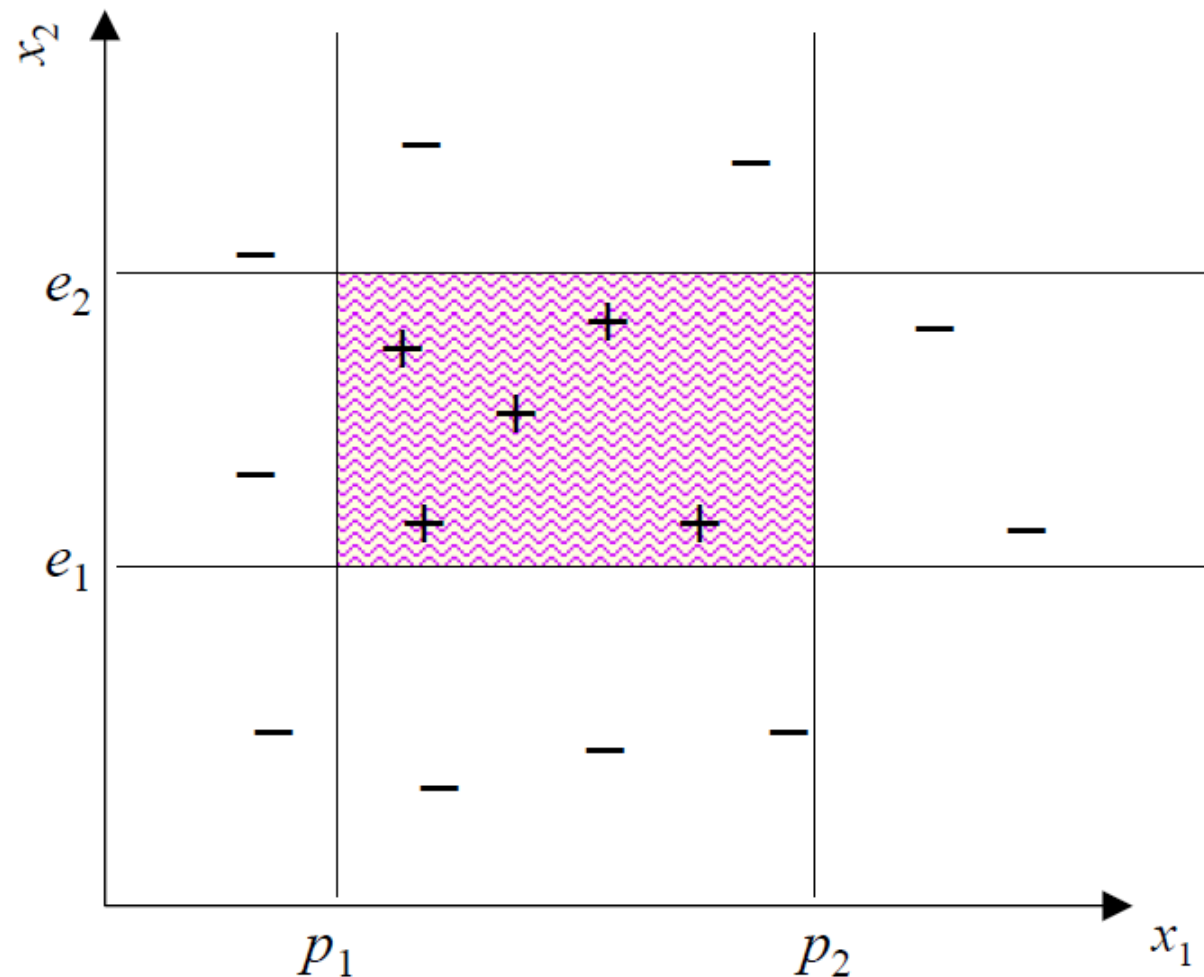
假设类

Hypothesis Class

- 假设我们认为，一辆汽车要想被归为跑车，其售价与发动机功率需要处于特定区间内：

suppose that we think that for a car to be a sports car, its price and its engine power should be in a certain range:

$$\begin{cases} p_1 \leq \text{price} \leq p_2 \\ e_1 \leq \text{engine} \leq e_2 \end{cases}$$

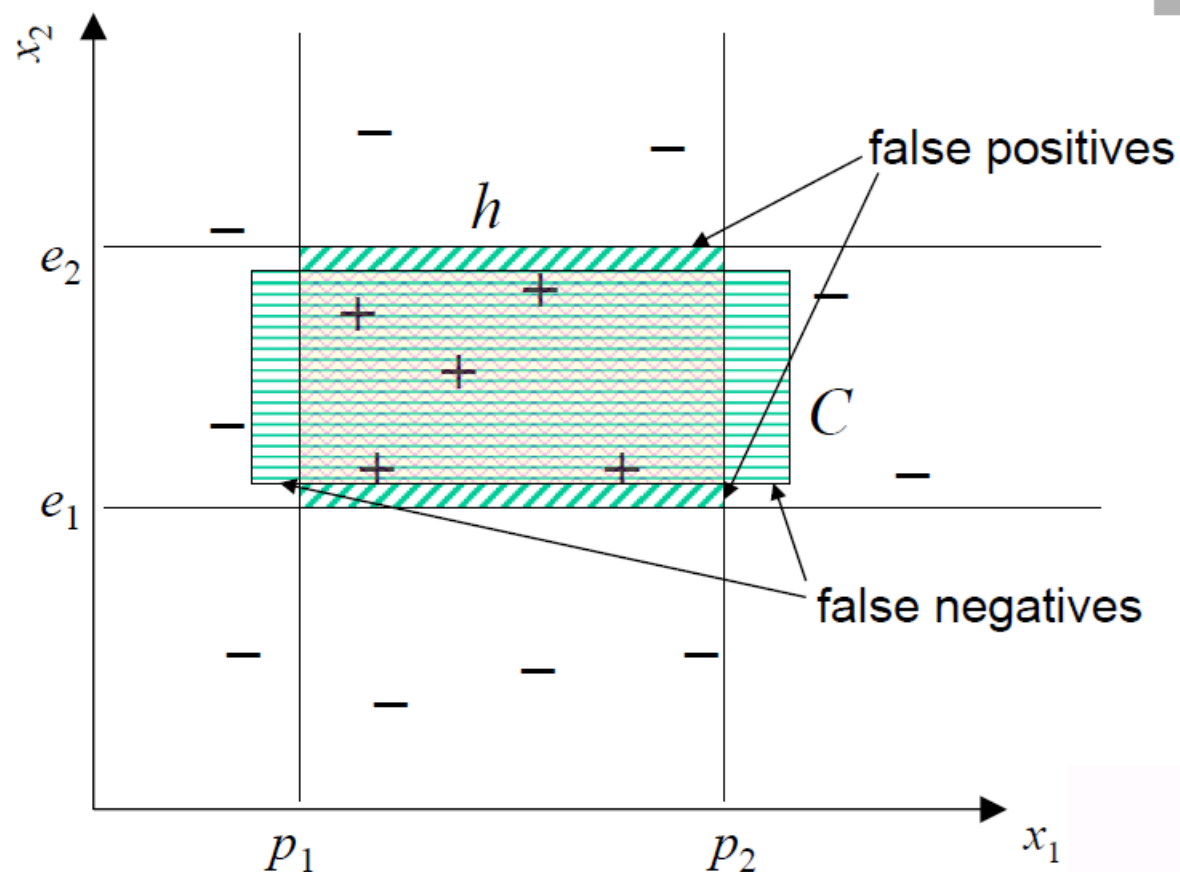


概念类

Concept Class

- 假设真实类别为 C ，机器学习的任务是从假设空间 $\mathcal{H}$ 中找到一个假设 h ，使其与样本集 $\mathcal{X}$ 保持一致。

Suppose that the actual class is C , the task of Machine Learning is to find $h \in \mathcal{H}$ that is consistent with $\mathcal{X}$.



假设选择

Hypothesis Choice

- 为了找到需要的假设 h ，我们需要定义经验误差的概念：假设 h 的预测结果与训练集真实标签不一致的样本所占的比例，即：

To find a hypothesis h , we need to define the empirical error: proportion of training instances where predictions of h do not match the training set, aka.,

$$E(h|X) = \frac{1}{N} \sum_{t=1}^N 1(h(\mathbf{x}^t) \neq y^t)$$

- 每一组 (p_1, p_2, e_1, e_2) 都定义了一个属于假设类 $\mathcal{H}$ 的假设 h 。机器学习的任务是从中找出最好的那一个假设。

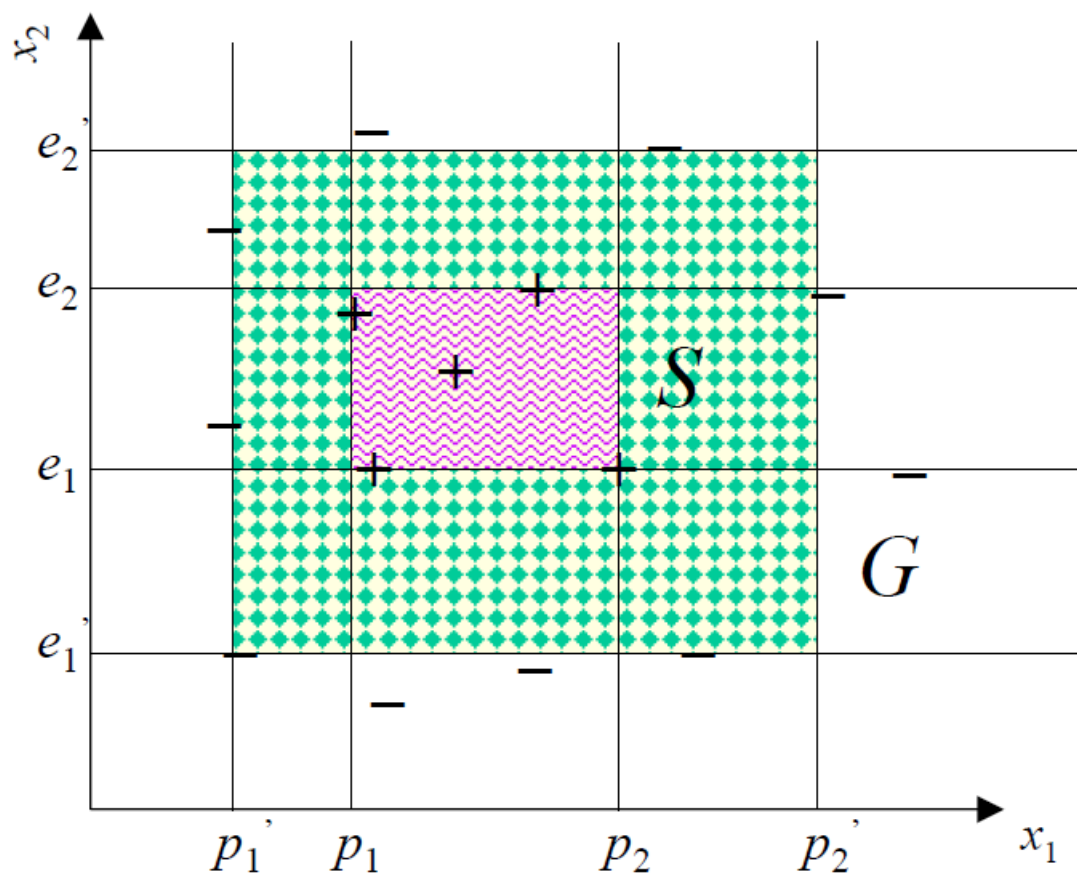
Each (p_1, p_2, e_1, e_2) defines a hypothesis $h \in \mathcal{H}$, the task of Machine Learning to find the best one.

假设选择

Hypothesis Choice

- ▶ 两个极限情况的假设如右图所示：
最宽泛的假设 g 与最严格的假设 s 。
 g : 如果边界能排除训练样本中的负样本，就是正样本的范围。
 s : 边界仅包含训练样本中的正样本。

The hypotheses for the two extreme cases are shown on the right: the most general hypothesis g and the most specific hypothesis s . g : If a boundary can exclude all negative examples in the training set, then it defines the range of positive examples. s : The boundary includes only the positive examples in the training set.

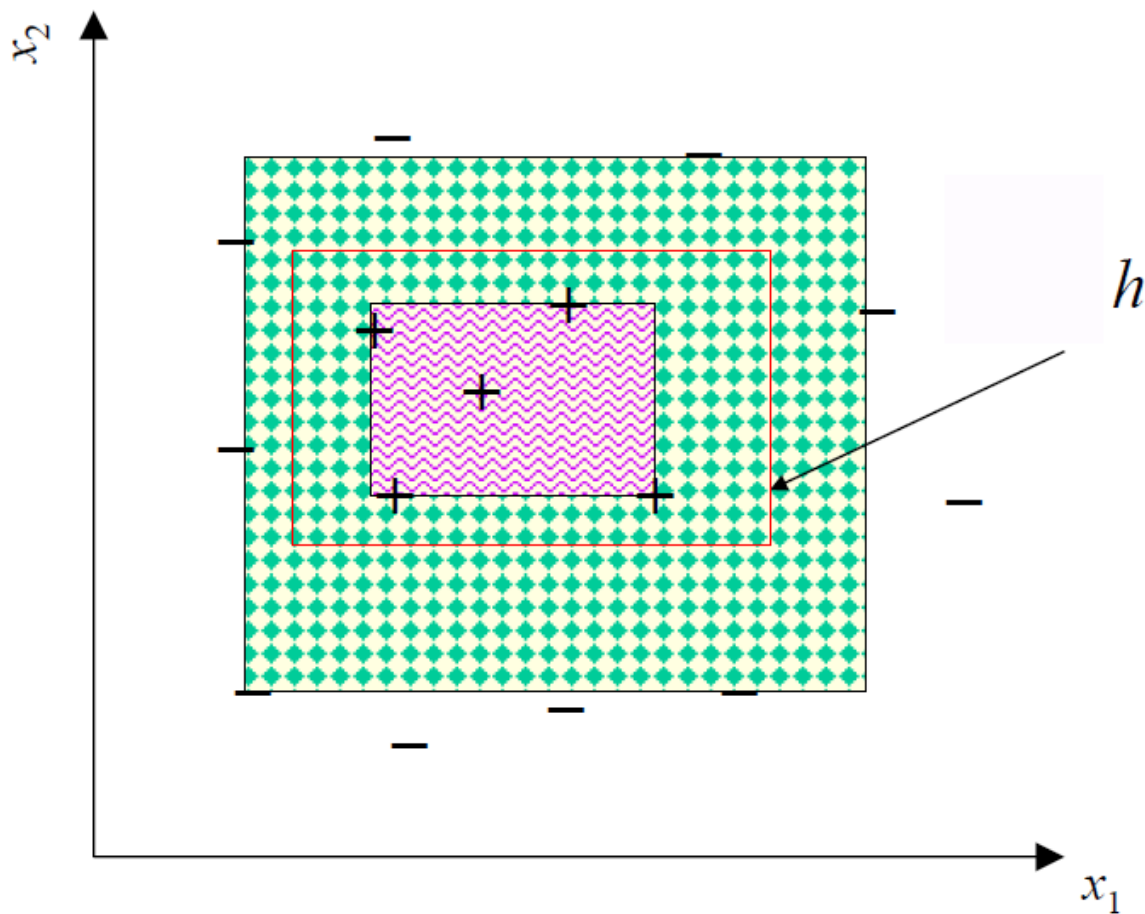


假设选择

Hypothesis Choice

- 所有与训练样本一致的假设的集合称为版本空间。从图中可以看出， g 和 s 定义了版本空间的边界，所有比 s 更宽泛、且比 g 更严格的假设所构成的集合，就是版本空间。

The set of consistent hypotheses forms the Version Space. From the figure, it is manifest that g and s define the boundaries of the Version Space. The set of hypotheses more general than s and more specific than g forms the Version Space.



机器学习

Machine Learning

- 假设：训练样本从一个未知但固定的概率分布 $P(\mathbf{x}, y)$ 中独立采样得到的。

Assumption: training instances are drawn from an unknown but fixed probability distribution $P(\mathbf{x}, y)$ independently.

- 学习任务：

- 给定训练集 $S = \{(\mathbf{x}^1, y^1), (\mathbf{x}^2, y^2), \dots, (\mathbf{x}^l, y^l)\}$;
- 希望构建一个规则 f ，预期将样本 $\mathbf{x}$ 映射到标签 y ;
- 如果 $f(\mathbf{x}) \neq y$ ，则会有相应的损失或惩罚，如 $\ell(f(\mathbf{x}), y) = |f(\mathbf{x}) - y|$ 。

- Learning Task:

- Given a training set $S = \{(\mathbf{x}^1, y^1), (\mathbf{x}^2, y^2), \dots, (\mathbf{x}^l, y^l)\}$;
- To construct a rule f , which can potentially predict the label y given $\mathbf{x}$;
- If $f(\mathbf{x}) \neq y$, there is some loss or penalty, for example, $\ell(f(\mathbf{x}), y) = |f(\mathbf{x}) - y|$.

Representation + Optimization + Evaluation

表示 + 优化 + 评估

机器学习

Machine Learning

► 在跑车与非跑车的例子中：

► 表示：

- 用两个属性来表示一辆汽车——价格和发动机排量；
- 用一个矩形来表示假设（即决策规则）。

► 优化：在某种正则化约束下，选择误差最小的假设。

► 评估：

- 我们的模型在预测未来样本时表现会好吗？
- 它会有多好？
- 是否存在其他设置可以让它表现得更好？

► In the sport cars vs. non-sport cars example:

► Representation:

- Represent a car by two attributes: price and engine volumn
- Represent he hypothesis (decision rule) by a rectangle

► Optimization: Choose the hypothesis with the smallest error under certain regularization.

► Evaluation:

- Will our model work well in predicting the future examples?
- How good it will be?
- Is there any different setting to make it perform better?